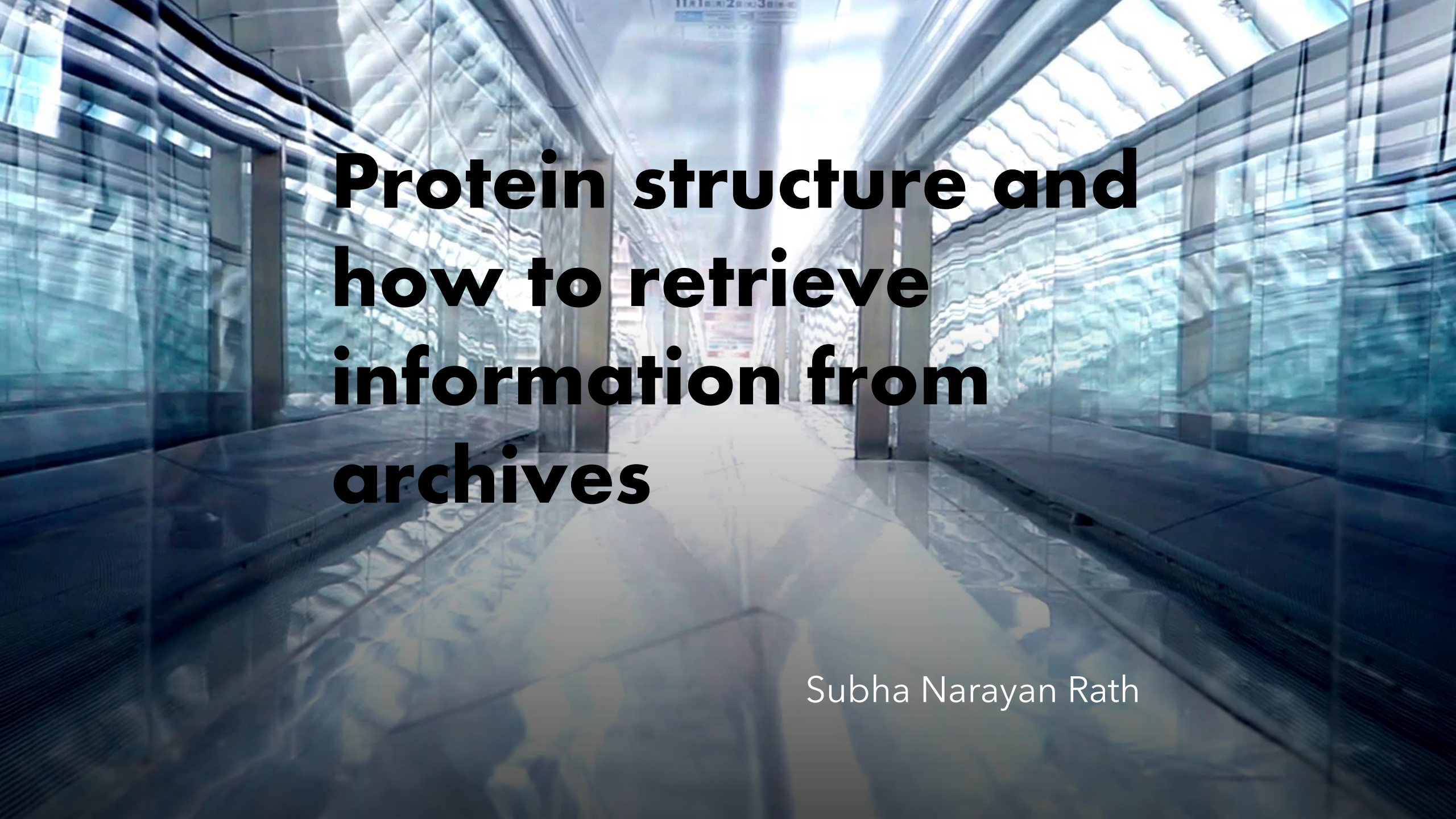


The background of the slide is a photograph of a modern subway station. The station has a high ceiling with a complex, ribbed structure and large glass walls that reflect the interior. The floor is polished and reflects the overhead lights. The overall color palette is dominated by blues and greys, giving it a clean, technological feel.

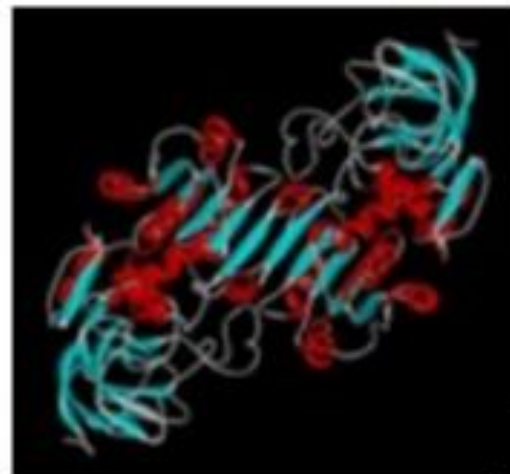
Protein structure and how to retrieve information from archives

Subha Narayan Rath

- Three examples of protein functions

- Catalysis:

Almost all chemical reactions in a living cell are catalyzed by protein enzymes.



Alcohol dehydrogenase oxidizes alcohol to aldehydes or ketones

- Transport:

Some proteins transport various substances, such as oxygen, ions, and so on.



Haemoglobin carries oxygen

- Information transfer:

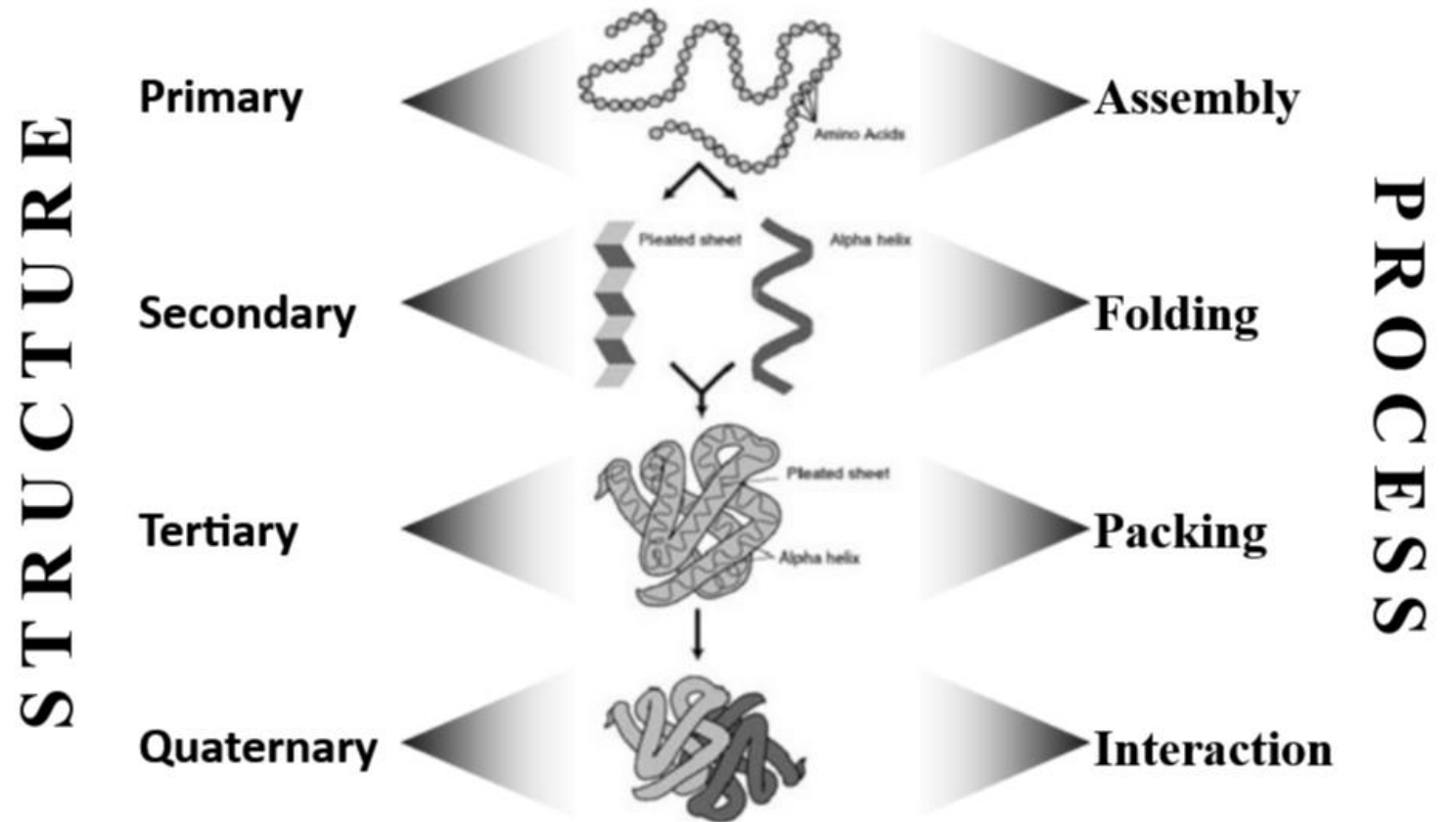
For example, hormones.



Insulin controls the amount of sugar in the blood

Basic unit: amino acids

- Depending on R sub-unit of amino acids the properties vary
- 20 amino acids and their properties
- Polymerization of amino acids while synthesis, causes the primary structure

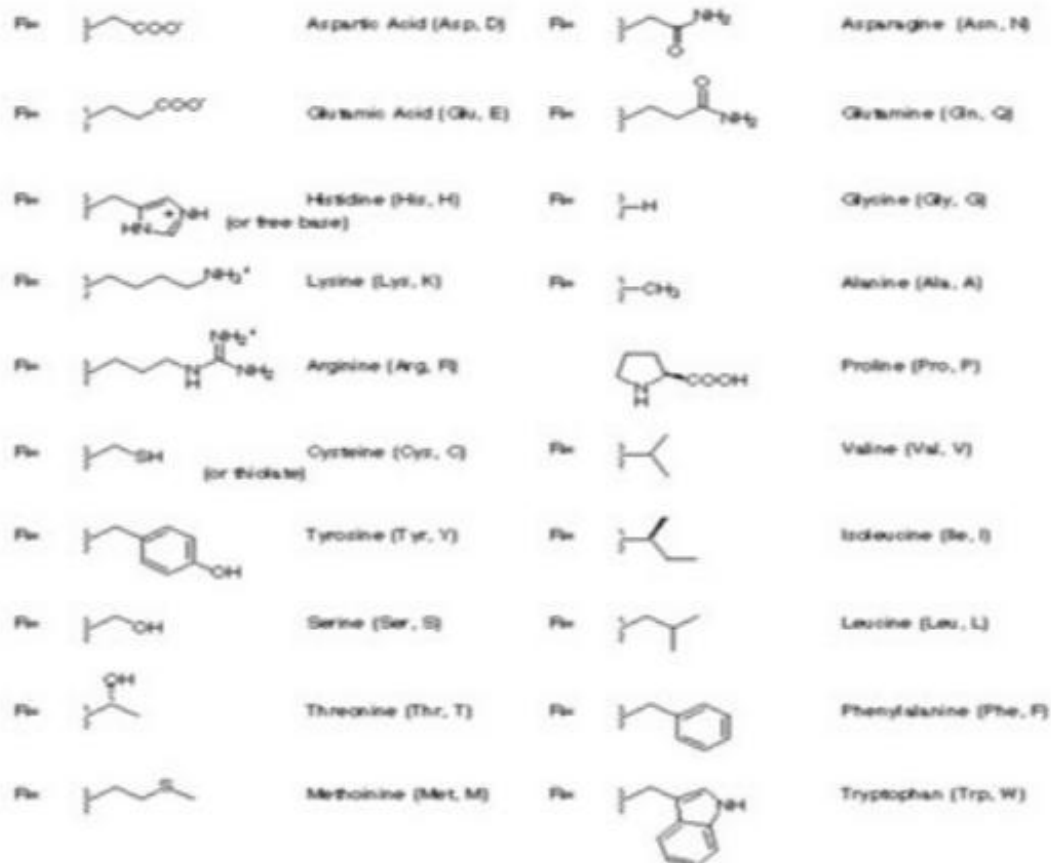


Primary structure

primary structure of human insulin

CHAIN 1: GIVEQ CCTSI CSLYQ LENYC N

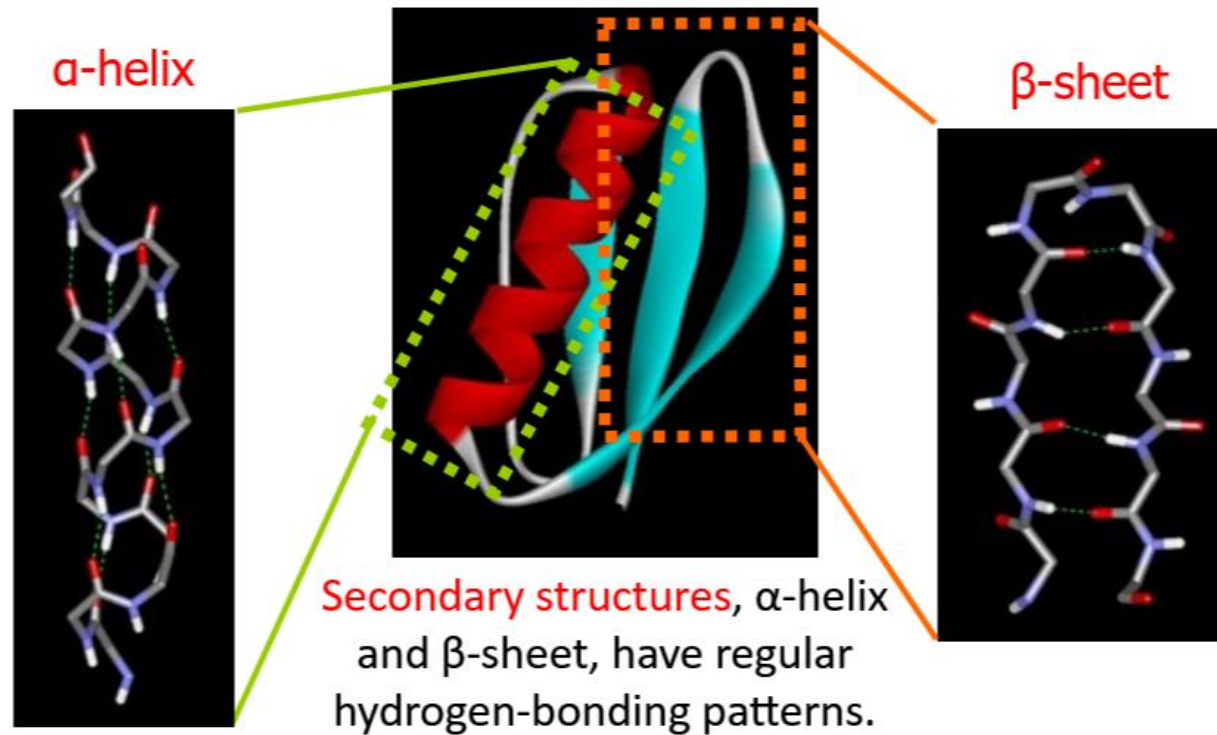
CHAIN 2: FVNQH LCGSH LVEAL YLVCG ERGFF YTPKT



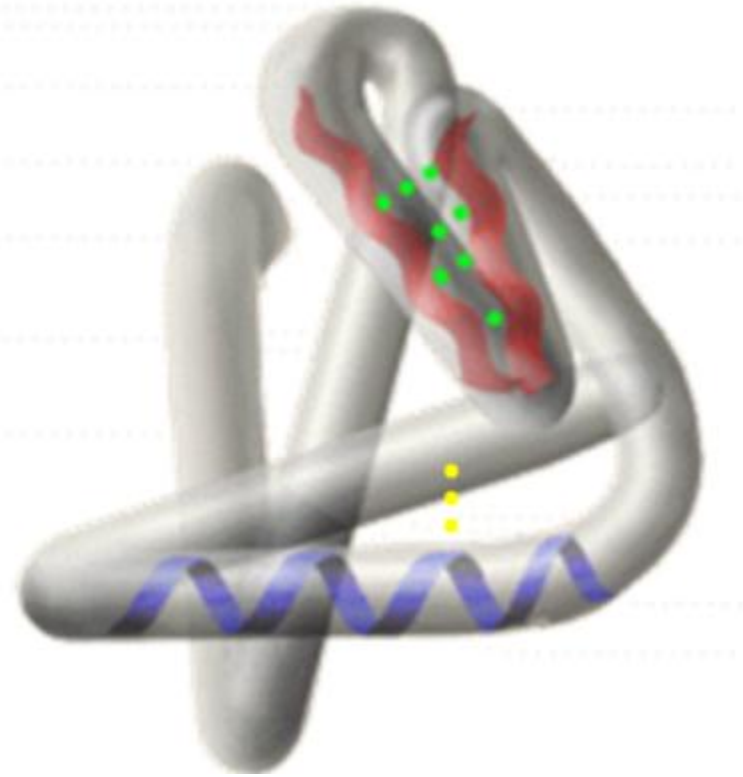
-
-
- Linear and ordered
- 1D
- Sequence of amino acids
- Written from amino end to carboxyl end by convention
- This linear structure is neither functional nor stable; so it has to do folding

Secondary structure due to protein folding

- Non-linear
- 3D
- Localized to regions of amino acid chain
- Formed and stabilized by
 - Hydrogen bonding
 - Van-der-walls interactions
 - Electrostatic forces

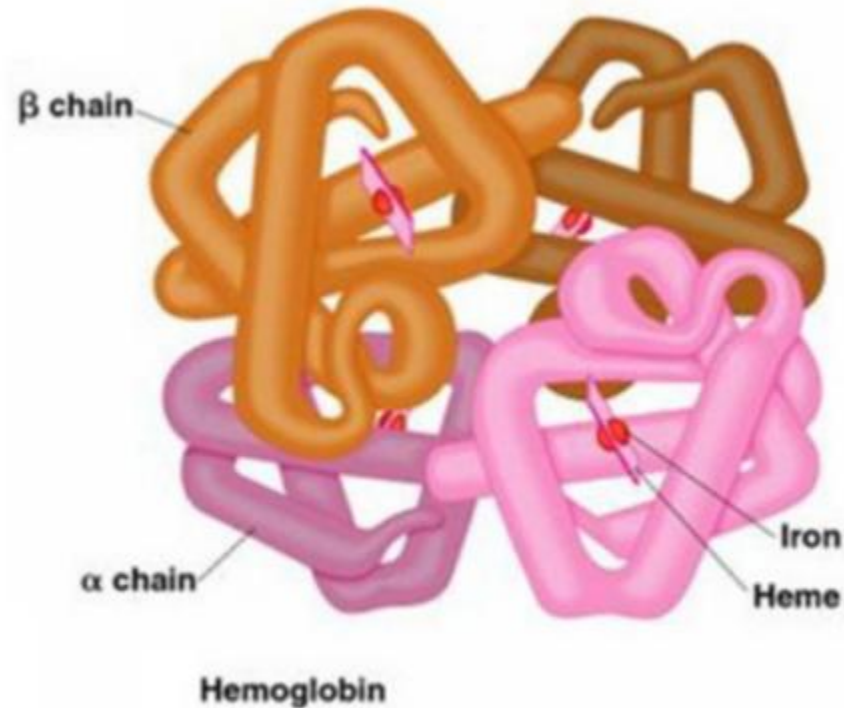


Tertiary structure due to protein packing



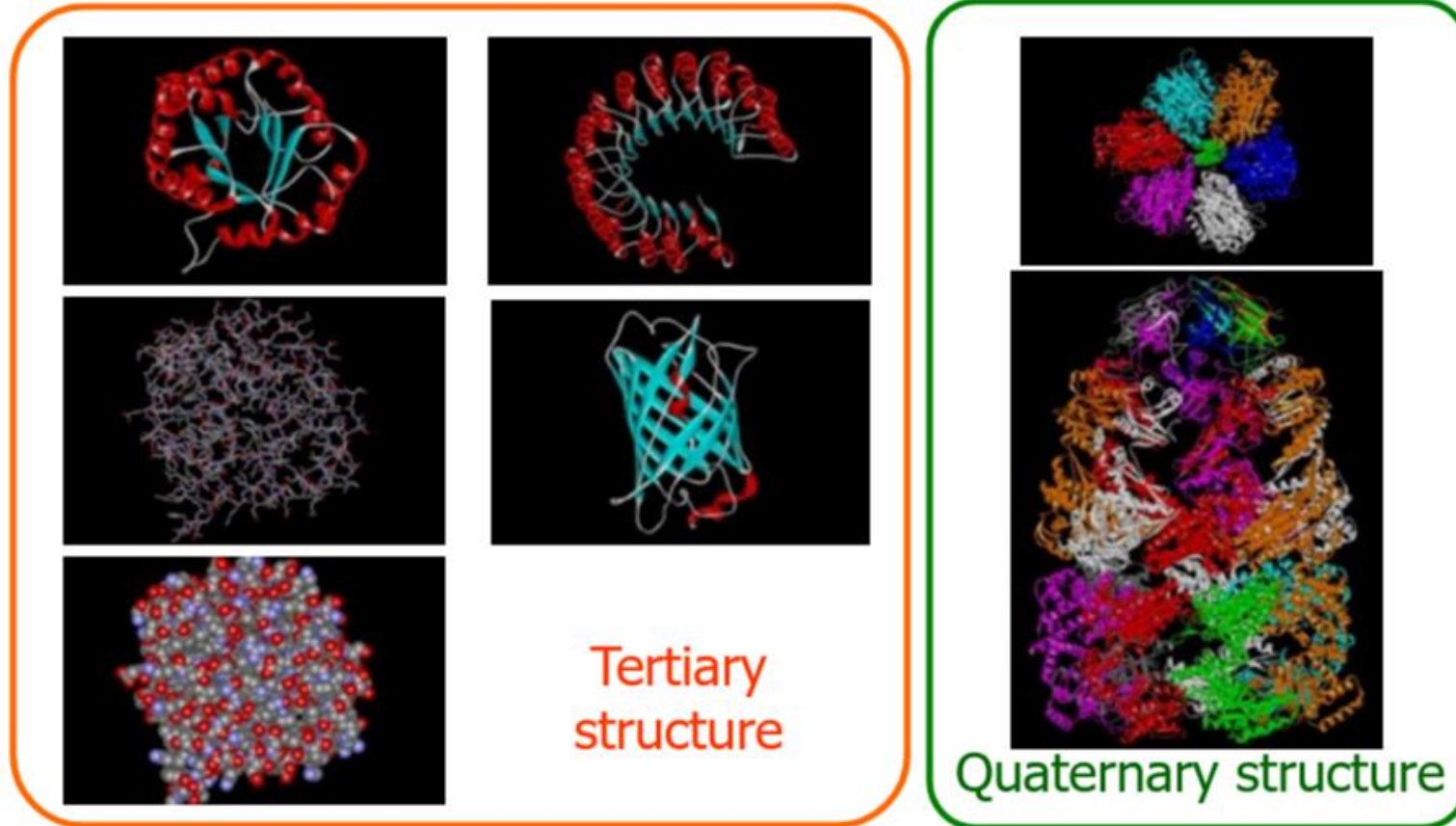
- Occurs in cytosol (upto 60% of bulk water or 40% of water of hydration)
- Due to interaction of 2 structure and solvents
- It's non-linear and 3D

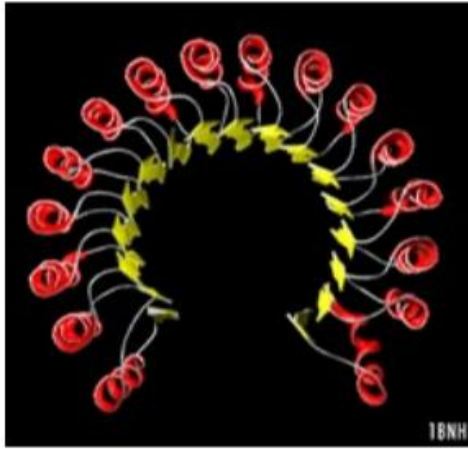
Quaternary structure due to protein interactions



- In cytosol due to close proximity to other folded and packed proteins
- Involve interaction of tertiary structure elements of separate protein molecules
- Non-linear and 3D

3d structure of proteins





α/β

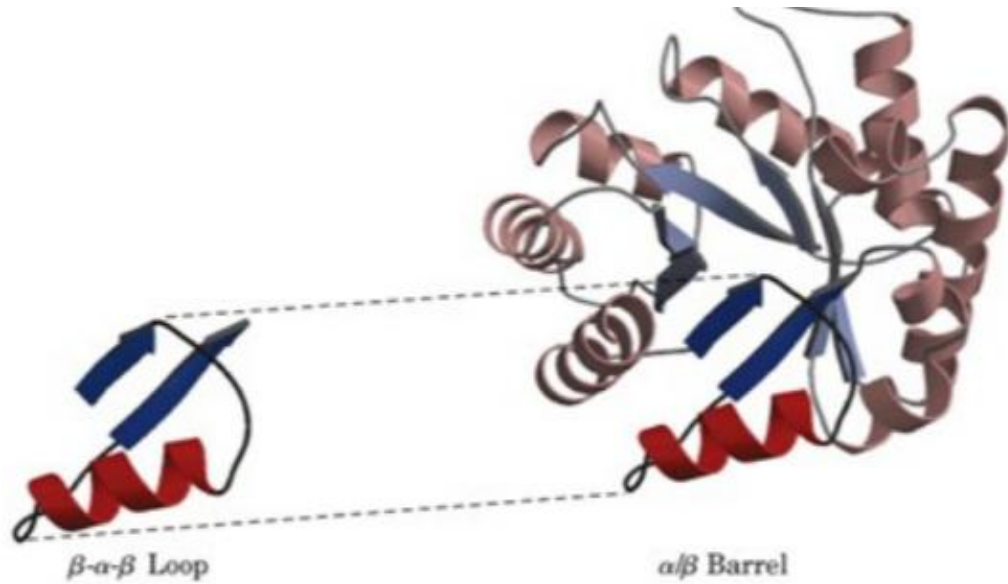


β - α - β Loop

Class/motif

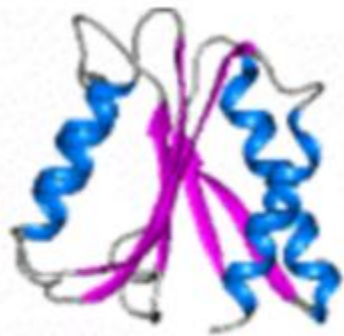
- Class = secondary structure composition
- Could be all α , all β , α/β , $\alpha+\beta$
- Motif = small specific combination of secondary structure elements e.g. β - α - β loop

Fold



- Fold is architecture, the overall shape and orientation of secondary structures, ignoring connectivity between the structures
- E.g. alpha/beta barrel

Fold families or superfamilies



flavodoxin
(4fxn)

CLASS: $\alpha+\beta$

FOLD: sandwich

FOLD FAMILY: flavodoxin

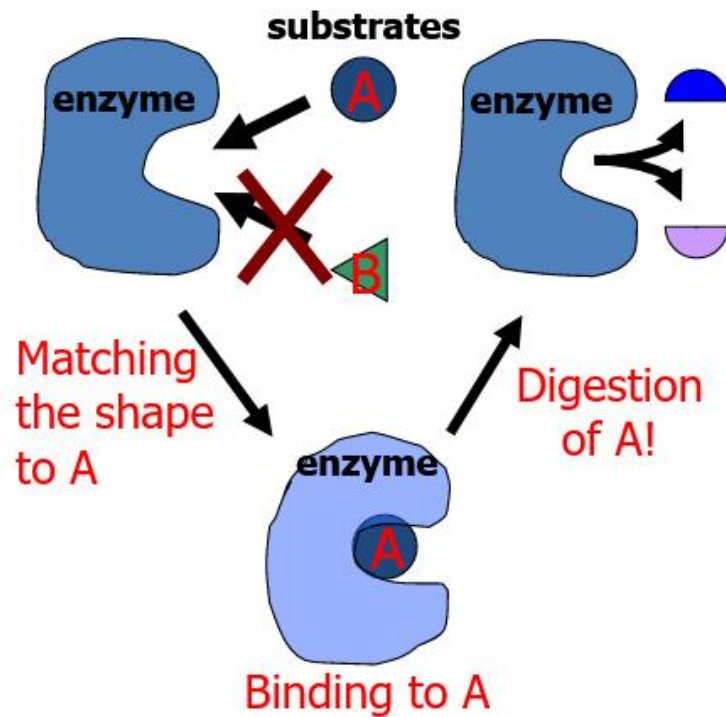
- Fold families: categorizations that take into account topology and previous subsets as well as empirical or biological properties
- Superfamilies: above plus it includes evolutionary and ancestral properties

Hierarchical protein structure

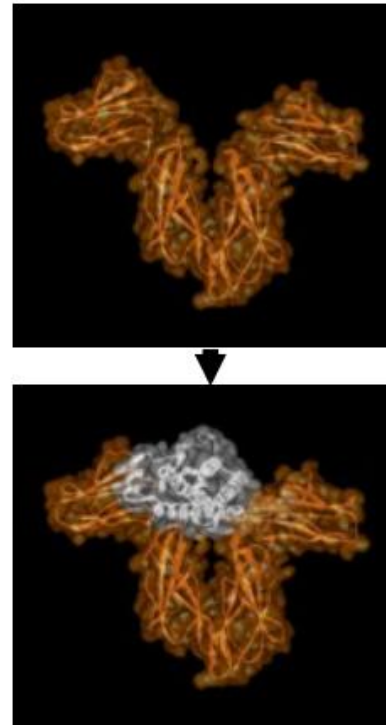
- Primary (amino acid sequence)
- Secondary (alpha helix and beta sheet)
- Tertiary (3D structure formed by assembly of secondary structures)
- Quaternary (more than one polypeptide chains)

Protein structure and function

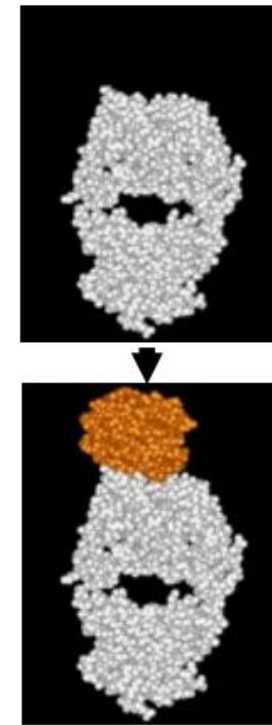
Example of enzyme reaction



Hormone receptor



Antibody



BLASTP is for:

- Exploring protein function
- Initial discovery for conserved domains
- However, can we predict protein structure and function based on amino acid sequence??
- Each protein folds to a unique 3D structure based on its amino acid sequence
- Protein structure is closely related to its function
- Protein structure prediction is a grand challenge of computational biology

Proteomics

- 2 methods can measure:
 - High resolution 2D PAGE
 - Mass spectrometric techniques
- Previously by direct sequencing of proteins; but now by translation of DNA sequences...new protein sequence data are determined; but the later is always hypothetical unless experimentally proven
- The pattern recognition programs that can do it will be subject to 3 types of errors:
 1. protein sequences might be missed entirely, may incorrectly spliced, might be from exons in different ways of combination and in different tissues can't be predicted. If mRNA is edited before translation, it can't be known.
 2. No clue for quarternary structure, prosthetic group binding, patterns of disulphide bridges
 3. Post translational modifications: covalent alterations within a cell, addition of a ligand, or cleavage of a protein to an active form
- Inteins are proteins that have self splicing activity compared to done by proteases

Protein sequence database

- 1. Protein information resource (PIR) and associated database
 - A. PIRSF: Sequence family classification
 - B. iProClass: integrated protein knowledgebase
 - C. iProLINK: integrated protein literature, information and knowledge
- 2. SWISS-PROT (from Swiss institute of bioinformatics, Geneva, Switzerland)
 - Also contains bioinformatics tools and links called Expert Protein Analysis System (ExPASy....www.expasy.org)
 - PROSITE is a set of signature patterns characteristic of protein families
- 3. TrEMBL

In 2002, all three of them coordinated to form: UniProtKB consortium

- Amino acid sequence is not inferable from the gene sequence, because
- A. ambiguity in splicing
- B. information about ligands, disulphide bridges, subunit associations, post-translational modifications, effects of mRNA editing etc. can't be known from gene sequence
- E.g. <https://www.uniprot.org/>

Database of protein families

- Patterns of conservation identity features that nature has found to retain (PROSITE signatures are examples)
- R. Doolittle suggested: two full length protein sequences (>100 residues) that have 25% or more identical residues in an optimal alignment are likely to be related
- < 15%; doubtful similarity
- 18-25%: twilight zone where there might be similarity like appearance of PROSITE consensus patterns
- Sequence oriented databases are: interPro, Pfam, COG
- Structure-oriented databases are: SCOP, CATH

DATABASE OF PROTEIN STRUCTURE

- They archive, annotate, distribute a set of atomic coordinates
- Worldwide Protein DataBank (wwPDB)
- Others: protein data bank Europe (EBI at UK) and protein data bank Japan (based at Osaka university)
- www.wwpdb.org
- It contains also structures of nucleic acids, carbohydrates in addition to proteins
- CCDC: Cambridge crystallographic Data Centre archives the structure of small molecules
- BioMagResBank at University of Wisconsin: archives protein structures determined by NMR
- wwPDB keeps the data from X-ray structure determinations

A protein databank structure contains:

- What protein and from which species
- Who solved the structure with reference
- Experimental details to solve the structure such as resolution of xray structure determination
- Amino acid sequence
- Atomic coordinates (starting with ATOM)
- Additional molecules including cofactors, inhibitors, water molecules (keyword HETATM identifies coordinates of these)
- Assignment of secondary structure: helices and sheets
- Disulphide bridges